CIS 385 – PROJECT MANAGEMENT · UNIVERSITY OF THE FRASER VALLEY · HIGH-LEVEL PLAN

UFV SkyGrid

Autonomous Aviation Research, Education & Regional Field-Test Network

"Where Autonomous Aviation Meets the Real World." · Project authorization ceiling: CAD $100,000,000 · Required launch: September 1, 2030 · Active assignment team: Anmol • Armaan • Ishmeet Anand

1. The idea

UFV SkyGrid is a proposed $100-million autonomous aviation research, education, and regional field-test network designed to connect UFV students and researchers with the Fraser Valley's aerospace, agriculture, technology, infrastructure, and public-sector ecosystems. Rather than creating only a conventional drone laboratory, SkyGrid would combine a central simulation and innovation capability with regulatory-compliant field-testing locations across the Fraser Valley, allowing autonomous aerial systems to be researched and evaluated under realistic operating conditions. The project would operate around three priorities: **Learn, Innovate, and Impact**.

The business problem: "UFV has relevant academic disciplines, applied-research ambitions, experiential-learning programs, and access to a region rich in autonomous-system applications, but those assets are not presently organized around a shared platform capable of moving work from classroom learning to simulation, field testing, and industry application."

Why now: In Abbotsford, agriculture alone includes approximately 1,254 farms operating across 61,586 acres and generating nearly $1 billion in annual farm-gate sales (City of Abbotsford, 2024). Transport Canada introduced a Level 1 Complex pathway for certain lower-risk BVLOS operations beginning November 4, 2025 (Transport Canada, 2026a). SkyGrid uses the Abbotsford aerospace ecosystem mainly as a knowledge, industry, simulation, and systems-integration anchor while locating real-world field testing only where the regulatory environment permits it.

2. High-level project charter

Project	UFV SkyGrid — Autonomous Aviation Research, Education & Regional Field-Test Network
Purpose	Create a university-wide autonomous-aviation platform connecting student learning, applied research, simulation, field testing, and regional industry/community challenges.
2030 vision	PROPOSED aspiration Establish UFV SkyGrid as Canada's leading university-based regional testbed for applied autonomous aviation.
Project owner	University of the Fraser Valley.
Executive sponsor	PROPOSED UFV President/Vice-Chancellor or Board-designated executive sponsor with authority appropriate to the investment.
Project manager	PROPOSED Dedicated SkyGrid Program Director / Project Manager.
Start	November 2026 following authorization.
Required completion	September 1, 2030.
Budget ceiling	CAD $100,000,000 including reserves and transition-to-operations funding.
Primary pillars	Learn → Innovate → Impact.
Proposed measurable launch targets	PROPOSED At least 3 compliant field-test environments; at least 8 substantive written external partnership agreements; participation pathways across at least 5 UFV academic/program areas; no unresolved critical safety, regulatory, cybersecurity, or privacy deficiencies.
Authorization principle	Continue only while demand, regulatory feasibility, site access, affordability, and partner participation remain credible. Formal gates permit Continue, Redirect, or Terminate / Limited-Launch decisions.

In scope

- Central research, simulation, and collaboration capability.
- Regulatory-compliant regional field-test locations.
- RPAS platforms, payloads, sensors, command-and-control, and simulation systems.
- Secure compute, data, networking, cybersecurity, and privacy controls.
- Academic and applied-research integration.
- Student applied-learning pathways.
- Industry/community challenge process and external partnerships.
- Staffing, operator readiness, safety management, regulatory compliance, insurance, commissioning, and handover.

Out of scope

- A commercial airline or conventional flight school.
- Replacement of Transport Canada, NAV CANADA, airport-authority, or other regulatory functions.
- Weaponized or offensive military drone development.
- Passenger-carrying autonomous-aircraft services.
- Unrestricted BVLOS operations outside approved Canadian regulatory pathways.
- Routine independent emergency-response operations by UFV.
- Mass manufacturing of certified aircraft.

3. Work breakdown structure (Level 2) and critical packages

- **1.1** Project Governance & Control
- **1.2** Feasibility & Investment Authorization
- **1.3** Innovation & Simulation Node
- **1.4** Fraser Valley Field-Test Network
- **1.5** Autonomous Systems & Technical Infrastructure
- **1.6** Academic, Research & Student Experience
- **1.7** External Partnership & Sustainability Model
- **1.8** Safety, Regulatory, Cybersecurity & Privacy System
- **1.9** Integration, Commissioning & Transition

Critical packages (WBS dictionary): **1.2.3** Regulatory & airspace feasibility; **1.3.7** Commissioned central node; **1.4.6** Commissioned field-test network; **1.5.1** Modular RPAS fleet; **1.6.2** Student Applied-Learning Program; **1.7.1** Strategic partner portfolio; **1.8.2** Safety Management System; **1.9.5** Operational Readiness Review. Each has documented acceptance evidence and a primary dependency.

4. Delivery approach — prove before we spend

Core control principle: SkyGrid does not receive an unconditional $100M spending commitment on Day 1. Each major investment stage must earn permission to proceed. Hybrid, stage-gated delivery: predictive methods for facilities, sites, capital procurement, regulatory deliverables, budgets and milestone dates; adaptive methods for software, analytics, prototypes, research use cases and partner pilots (PMI, n.d.).

PHASE	PERIOD	DECISION / OUTPUT
1. Authorization & Mobilization	Nov–Dec 2026	Project formally initiated.
2. Feasibility & Demand Validation	Jan–Jun 2027	Gate 1 — Prove Need.
3. Site, Regulatory & Concept Definition	Jul–Dec 2027	Gate 2 — Prove Permission.
4. Detailed Design & Procurement Readiness	Jan–Jun 2028	Gate 3 — Prove Affordability.
5. Build, Procure & Develop	Jul 2028–Dec 2029	Gate 4 — Prove Capability.
6. Integration, Pilots & Readiness	Jan–Jun 2030	Gate 5 — Prove Readiness.
7. Final Acceptance & Launch	Jul–Sep 1 2030	Operational launch Sep. 1, 2030.

The five gates are kill points: each permits Continue, Redirect, or Terminate / Limited-Launch decisions. Example — Gate 3 — Capital investment: continue only if design mature enough for reliable costing; total authorization ≤$100M incl. reserves; site agreements advanced; regulatory path viable; tech specs fit intended missions.

The external deadline remains September 1, 2030; the proposed internal operational-acceptance target is August 15, 2030. The full Gantt chart (Table 10, 24 activities and milestones M1–M7) is attached.

5. $100M cost baseline Proposed

Financial principle: the $100M is an authorization ceiling, not a spending target.

BUDGET LAYER	AMOUNT
Planned SkyGrid work	$85M
Contingency reserve	$10M
Cost baseline	**$95M**
Management reserve	$5M
Maximum authorization	**$100M**

$85M PLANNED-WORK ALLOCATION	CAD M
Governance / feasibility / design	$6M
Innovation & Simulation Node	$22M
Field-Test Network	$10M
RPAS / sensors / equipment	$14M
Simulation / compute / data	$9M
Safety / regulatory / cyber / privacy	$5M
Academic / research / students	$7M
Startup staffing / capability	$6M
Partnerships / pilots	$3M
Integration / commissioning	$3M

Exact headcount and the hiring plan are deliberately deferred: staffing is covered at a functional level ($6M startup staffing/capability) and exact headcount is determined from workload.

6. Cost and change control

Funding is progressively released through stage gates. Each major WBS branch becomes a control account (PMI, 2026a). If forecast cost exceeds authorization: use valid contingency → value engineer → remove optional Category-C scope → defer enhancements → restructure/recompete procurement → redesign. Do not cut safety, required compliance, cybersecurity, or core student/research capability merely to preserve optional features.

CHANGE LEVEL PROPOSED	THRESHOLD	AUTHORITY
1 — Minor	Within approved work package; ≤$100K; no material scope/milestone/regulatory effect.	Program Director.
2 — Moderate	$100K–$1M or ≤20 working-day impact, with no change to total authorization or final deadline.	SkyGrid Change Control Board.
3 — Major	>$1M, critical-path impact, major scope/technical change, or significant contingency use.	Executive Sponsor / Steering Committee.
4 — Strategic	Changes $100M authorization, Sep. 1 deadline, fundamental mission, or major regulatory posture.	UFV senior governance / funding authority as applicable.

7. Top risks Proposed scores

ID	RISK	P×I	OWNER	PRIMARY RESPONSE
R3	Facility/infrastructure cost escalation	20 Critical	Finance + Program Director	Independent estimates; contingency; phased scope; value engineering.
R4	Technology purchased too early becomes obsolete	16 Critical	Technology Lead	Modular architecture; wave procurement; open interfaces where feasible.
R1	Regulatory pathway / approval delay	15 High	Regulatory Lead	Early Transport Canada engagement; alternate operating concepts.
R2	Suitable node/field sites not secured	15 High	Site/Capital Lead	Multiple site options; conditional agreements before capital release.
R7	Cyberattack compromises C2 systems or data	15 High	Cybersecurity Lead	Threat risk assessment; segmentation; secure C2; access/logging/response.

Safety-critical risks receive mandatory controls regardless of arithmetic score. Full register: 12 risks, every one owned.

8. Success

Proposed measurable launch targets: at least 3 compliant field-test environments; at least 8 substantive written external partnership agreements; participation pathways across at least 5 UFV academic/program areas; no unresolved critical safety, regulatory, cybersecurity, or privacy deficiencies. **Proposed North Star:** number of UFV students annually completing a field-validated or external-partner SkyGrid experience.

9. Team roles and responsibilities

STUDENT	ASSIGNMENT ROLE	PRIMARY RESPONSIBILITIES
Anmol	Project Integration & Project-Management Lead	Overall concept; charter integration; scope; WBS; delivery approach; gates/kill points; Gantt logic; consistency across all sections; final integration.
Armaan	Research, Stakeholder & Competitive-Analysis Lead	External research; stakeholder evidence; regulatory/environment scan; UFV/Fraser Valley evidence; competitor benchmarking; source verification.
Ishmeet Anand	Financial, Risk & Benefits Lead	$100M structure; sustainability; risk register; cost/change control; KPIs; benefits-realization logic.

PROPOSED / TEAM DESIGN: Numbers such as the three-site target, eight-partner target, five-program target, budget allocation, change thresholds, risk scores, and KPI targets are SkyGrid planning choices created by the team. They are not presented as commitments made by UFV, government, regulators, or external partners. Full APA references are on the website's Evidence page.